

WEEK 2 - ADVANCED NETWORKING

TASK 1 - Port & Services

Port :

A transport-layer number used to identify a network communication endpoint/service.

IP → Host

Port → Service/Endpoint

TCP/UDP

TCP - Connection-oriented, reliable

UDP - Connectionless, lower oriented

Important Port Associations

21 - FTP

22 - SSH

23 - Telnet

25 - SMTP

53 - DNS

67 - DHCP Server

68 - DHCP Client

80 - HTTP

110 - POP3

123 - NTP

143 - IMAP

161 - SNMP

389 - LDAP

443 - HTTPS

445 - SMB

514 - Syslog

587 - SMTP Submission

636 - LDAPS

993 - IMAPS

995 - POP3S

3389 - RDP

Key Point

Ports help identify network services, while TCP/UDP identifies the transport protocol used for the communication.

GROUP 1

Port Number	Protocol	Service Name	Usage
21	TCP	FTP	FTP control/commands
22	TCP	SSH	Secure remote login/administration
23	TCP	Telnet	Remote login
25	TCP	SMTP	Sending/relaying email
53	UDP/TCP	DNS	Domain name resolution

GROUP 2

Port Number	Protocol	Service Name	Usage
67	UDP	DHCP Server	DHCP server communication
68	UDP	DHCP Client	DHCP client communication
69	UDP	TFTP	Simple File Transfer
80	TCP	HTTP	Web Communication
110	TCP	POP3	Receiving email
123	UDP	NTP	Time synchronization

GROUP 3

Port	Protocol	Service	Usage
143	TCP	IMAP	Email access
161	UDP	SNMP	Network management/monitoring
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	TLS-protected web communication
445	TCP	SMB	Windows file/printer sharing
514	UDP	Syslog	Log/message transport
587	TCP	SMTP Submission	Email submission
993	TCP	IMAPS	IMAP over TLS
3389	TCP/UDP	RDP	Remote Desktop

TASK 2 - NETWORKING DEVICES

Router

What it is :

A device that connects different networks and forwards IP packets between them.

Main concepts:

Layer 3

IP addresses

Packets

Routing

Example: Home router connecting a private LAN to the internet.

Switch

What it is:

A device that connects devices within a LAN and forwards Ethernet frames using MAC addresses.

Main concepts:

Layer 2
MAC addresses
Frames
Local connectivity

Example: Office switch connecting PCs and printers.

Firewall**What it is:**

A security control that monitors and controls network traffic according to defined security rules.

Main concept:

Traffic inspection
Security rules
Allow/Deny

Example: Company firewall controlling traffic between the internet and internal network.

Router vs Switch**Router**

→ Different networks
→ IP
→ Packets
→ Routing

Switch

→ Same/local network
→ MAC
→ Frames
→ Local connectivity

Key memory

Switch connects. Router routes. Firewall controls.

TASK 3 - NETWORK SECURITY BASICS

MITM - Man In The Middle Attack

Definition

A MITM attack occurs when an attacker positions themselves between two communicating parties and may observe or interfere with their communication.

Concept



Real-World Example

Public Wi-Fi

An attacker may control or position themselves between a victim and the intended network destination.

Risks

- Data theft
- Password capture
- Communication interception/manipulation

Prevention

- HTTPS
- VPN
- Secure/trusted networks

Important Distinction

MITM = attacker positioned in the communication path.

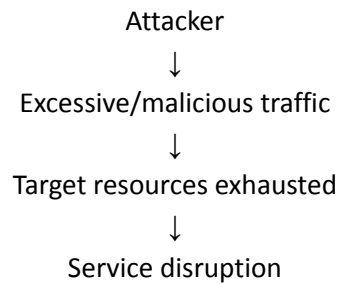
Packet sniffing = observing/capturing network traffic.

DoS - Denial of Service

Definition

A DoS attack attempts to make a system, service, or network resource unavailable to legitimate users.

Concept



Main Impact

Availability

Risks

- Service unavailability
- Performance degradation
- Resource exhaustion
- Legitimate users unable to access the service.

Basic Prevention

- Traffic filtering
- Rate limiting
- Firewalls
- Network monitoring

Key Point

DoS attacks primarily target the availability of systems or services.

PACKET SNIFFING

Definition

Packet sniffing is the process of capturing and examining network packets traveling across a network.

Concept

Network traffic

↓

Packets

↓

Captured

↓

Examined

Information that may be observed

- Source IP
- Destination IP
- Protocol
- Ports
- Packet size/timing
- Application information depending on protocol/encryption

Risks

- Sensitive information exposure
- Credential exposure
- Network information gathering

Prevention

- HTTPS/encryption
- VPN
- Secure/trusted networks

Difference from MITM

Packet sniffing focuses on capturing/observing traffic.

MITM involves positioning an attacker between communicating parties.

Key Point

Encrypted traffic can still be captured, but encryption protects the contents from being easily read by an observer.

PART 2 - PRACTICAL TASKS

TASK 4 - Packet Analysis

Wireshark is a network protocol analyzer used to capture and inspect network packets.

It helps identify:

- Source IP
- Destination IP
- Protocol
- Port numbers
- Protocol-specific details

A packet can be analyzed by selecting it and examining its IP, TCP/UDP, and application-protocol details.

Practical Observation

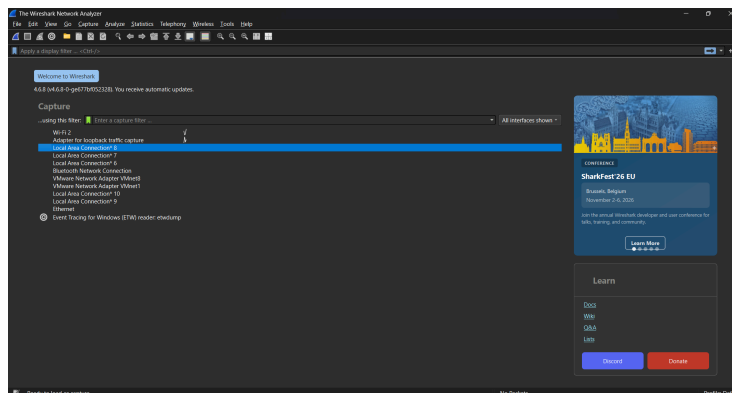
Command/Filter: **dns**

Observation: A DNS response was identified using UDP port 53. The selected packet showed source IP **10.245.23.130** and destination IP **10.245.23.59**.

Filter: **tls**

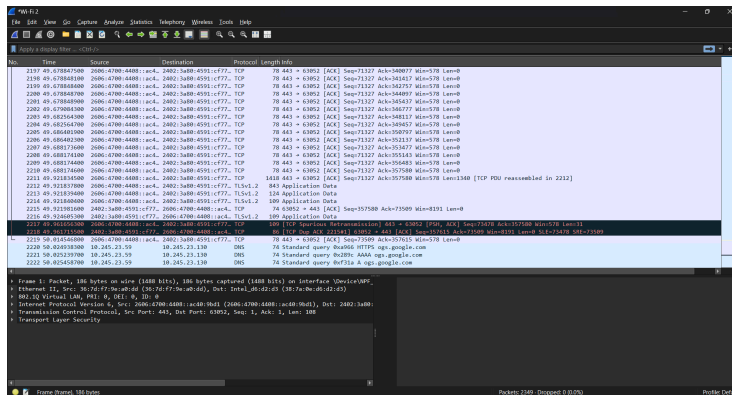
Observation: A TLSv1.2 application-data packet was identified. The selected connection used source port **443**, showing TLS-protected web communication.

Step 1 - Select the correct interface



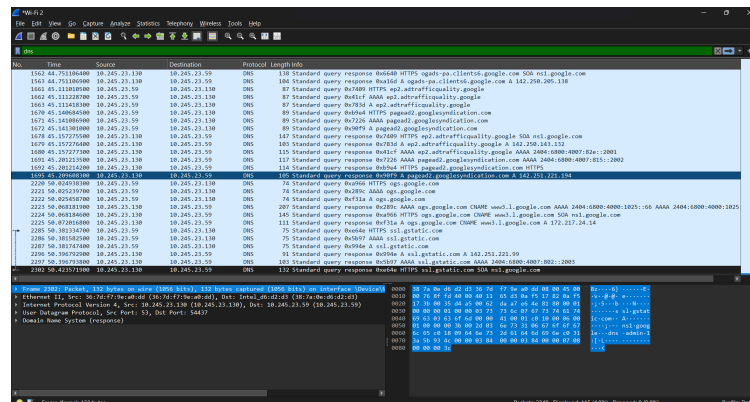
Selected the active Wi-Fi 2 interface in Wireshark and started packet capture.

Step 2 - Generate your own traffic



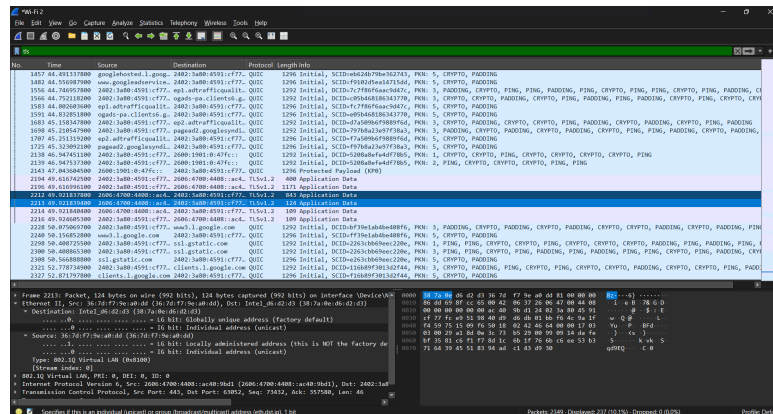
While capturing packets, I opened the browser and visited <https://google.com> to generate normal web traffic.

Step 3 - DNS Packet analysis



Applied the **dns** display filter and inspected a DNS response generated during the browsing session. The selected packet used UDP port 53, with source IP **10.245.23.130** and destination IP **10.245.23.59**.

Step 4 – HTTPS/TLS Traffic Analysis



Applied the **tls** display filter and inspected a TLSv1.2 application-data packet from the browsing session. The selected connection used source port **443**, demonstrating TLS-protected web communication.

TASK 5 - Network Scanning

Nmap (Network Mapper) is a network-scanning tool used to identify reachable ports, their states, and associated services on an authorized target.

Port → State → Service

Port States:

- **Open** - A service is accepting connections.
- **Closed** - No service is listening.
- **Filtered** - The port state cannot be determined because traffic is being filtered or blocked.

The **-sV** option is used to identify additional service and version information.

Commands:

nmap scanme.nmap.org

→ Basic port and service scan.

Nmap -sV scanme.nmap.org

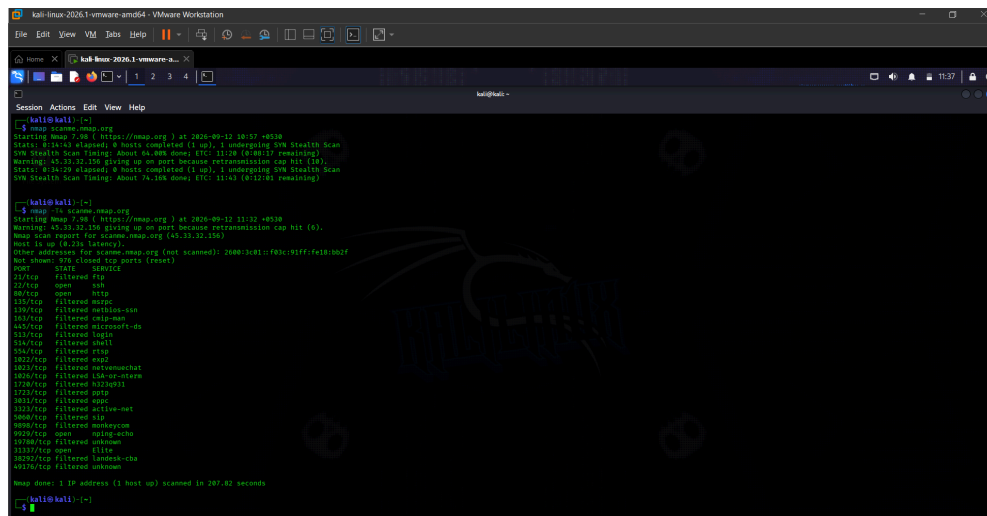
→ Attempts to identify service and version information.

Practical 1 - Basic Nmap Scan

Command:

nmap -T4 scanme.nmap.org

Executed the basic nmap scan on the authorized target scanme.nmap.org to identify reachable ports, their states, and associated services.



```
kali@kali:~$ nmap scanme.nmap.org
Starting Nmap 7.90 ( https://nmap.org ) at 2020-09-12 10:57 +0530
Nmap scan report for scanme.nmap.org (45.33.32.150)
Host is up (0.10s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    open     http
9929/tcp  open     nping-echo
31337/tcp open     elite
4532/tcp  filtered unknown
4537/tcp  filtered unknown
```

Practical Observation:

Command: **nmap -T4 scanme.nmap.org**

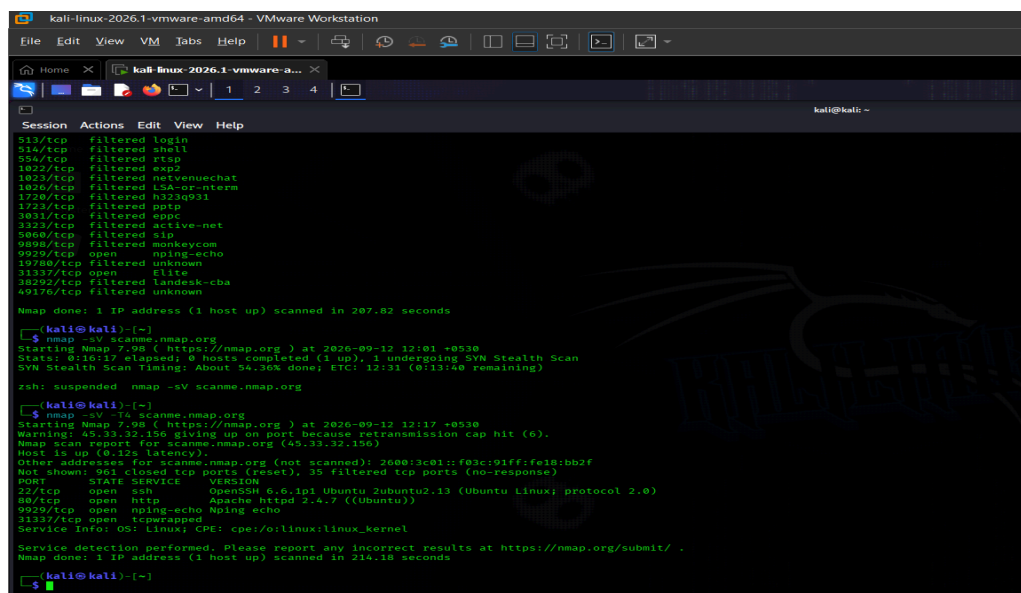
The scan identified scanme.nmap.org as an active host. Ports 22/tcp (SSH), 80/tcp (HTTP), 9929/tcp (nping-echo), and 31337/tcp (Elite) were found open, while several other ports were reported as filtered.

Practical 2 - Service Version Detection

Command:

```
nmap -sV -T4 scanme.nmap.org
```

Used the **-sV** option to identify additional service and version information for the discovered ports on the authorized target.



```
kali@kali: ~  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-12 12:01 +0530  
Stats: 0:16:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 54.36% done; ETC: 12:31 (0:13:40 remaining)  
zsh: suspended nmap -sV scanme.nmap.org  
[kali@kali] (~)  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-12 12:17 +0530  
Warning: 49.32.22.156 giving up on port because retransmission cap hit (6).  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.12s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c:91ff:fef0:bb2f  
Not shown: 961 closed tcp ports (reset), 35 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))  
9929/tcp  open  nping-echo   nping echo  
31337/tcp open  tcpwrapped  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 216.18 seconds  
[kali@kali] (~)
```

Practical Observation

Command: **nmap -sV -T4 scanme.nmap.org**

The scan identified the target as an active Linux host and detected four open TCP ports: 22 (SSH), 80 (HTTP), 9929 (nping-echo), and 31337 (tcpwrapped). Service/version detection identified OpenSSH 6.6.1p1 on port 22 and Apache httpd 2.4.7 on port 80.

Note: The official assignment command **nmap scanme.nmap.org** was attempted; however, the scan took an unusually long time to complete in my environment due to repeated retransmissions and did not complete within a reasonable period. Therefore, **-T4** timing optimization was used to complete the scan efficiently while keeping the same authorized target. The service/version detection was performed using **nmap -sV -T4 scanme.nmap.org**

TASK 6 - Packet Tracer LAN Setup

A LAN connects devices within a local network. A switch provides Layer 2 connectivity between devices and uses MAC addresses to forward Ethernet frames.

Both PCs were configured in the same /24 subnet, allowing them to communicate directly through the switch without a router.

IP Configuration

Device	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.11	255.255.255.0

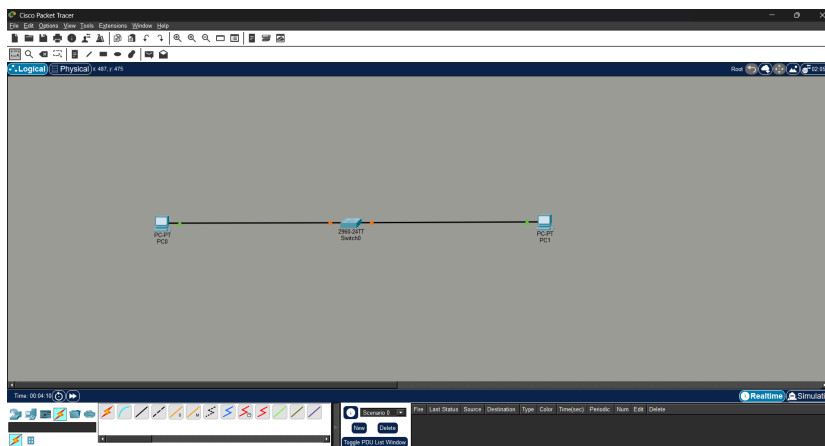
Practical Observation

PC1 successfully pinged PC2 (192.168.1.10) through the switch. All 4 packets were received with 0% packet loss, confirming successful communication between the two PCs on the same LAN.

Evidence

1. LAN Topology

PC1 → Switch → PC2



2. PC1 IP Configuration

192.168.1.10 / 255.255.255.0

The screenshot shows the configuration window for PC0. The 'Config' tab is selected, and the 'IP Configuration' section is active. The interface is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IP Address is '192.168.1.10', Subnet Mask is '255.255.255.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Automatic' is selected. The IPv6 Address is empty, Link Local Address is 'FE80::201:42FF:FE2B:8190', Default Gateway is empty, and DNS Server is empty. Under '802.1X', 'Use 802.1X Security' is checked. Authentication is 'MD5', Username is empty, and Password is empty. A 'Top' button is at the bottom left.

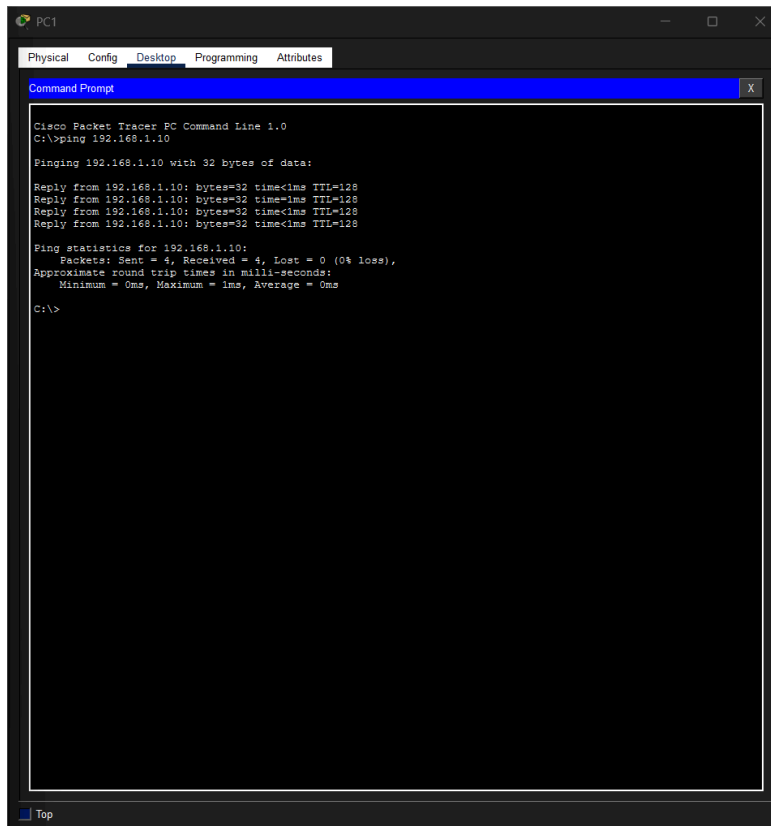
3. PC2 IP Configuration

192.168.1.11 / 255.255.255.0

The screenshot shows the configuration window for PC1. The 'Config' tab is selected, and the 'IP Configuration' section is active. The interface is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IP Address is '192.168.1.11', Subnet Mask is '255.255.255.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Automatic' is selected. The IPv6 Address is empty, Link Local Address is 'FE80::201:63FF:FE42:626E', Default Gateway is empty, and DNS Server is empty. Under '802.1X', 'Use 802.1X Security' is checked. Authentication is 'MD5', Username is empty, and Password is empty. A 'Top' button is at the bottom left.

4. Ping Result

ping 192.168.1.10 → 4 replies, 0% loss.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

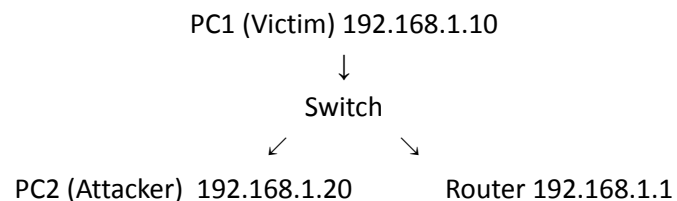
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

TASK 7 - Safe MITM Concept Simulation

A Man-in-the-Middle (MITM) situation occurs when an attacker positions themselves between two communicating parties and may potentially observe or interfere with their communication. The simulation uses a safe Packet Tracer topology to understand the attacker's conceptual position without performing real interception or credential theft.

Topology



Real-World Example

A public Wi-Fi network can create a risk of MITM attacks if an attacker is able to position themselves between a user's device and the intended destination.

Risks

- Data theft
- Password capture

Prevention

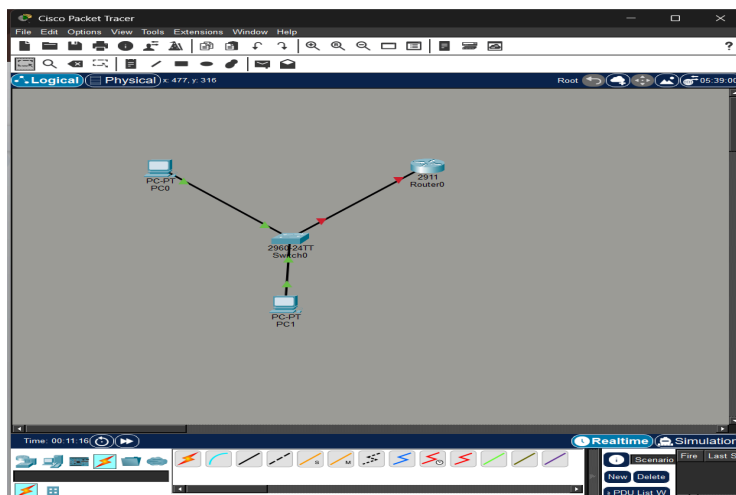
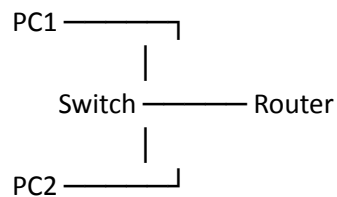
- HTTPS
- VPN
- Secure/trusted networks

Practical Setup

Step 1 - Create the Topology

Devices: PC1 (Victim), PC2 (Attacker), Switch, Router

Topology:



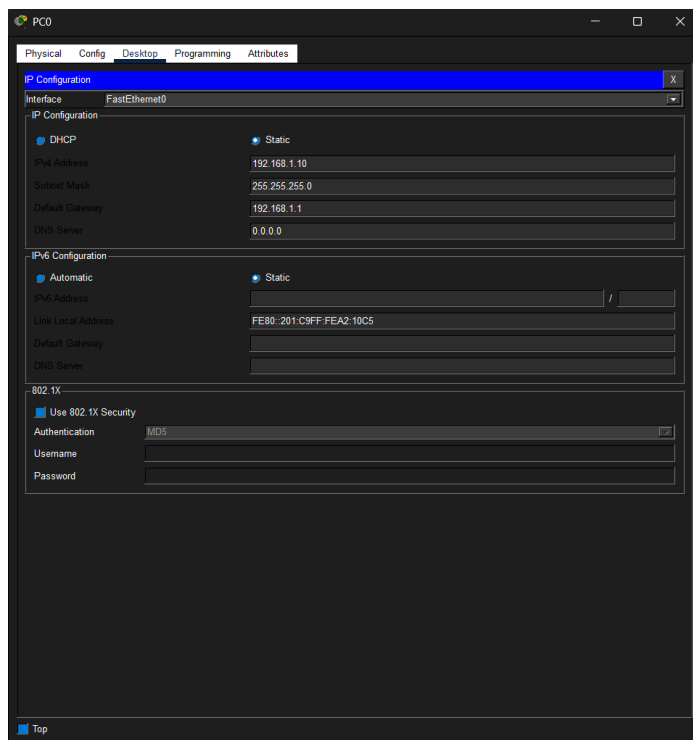
Explanation:

Created the safe Packet Tracer topology containing two PCs, a switch, and a router to represent the victim, conceptual attacker, and network path.

Step 2 - Configure IP Address

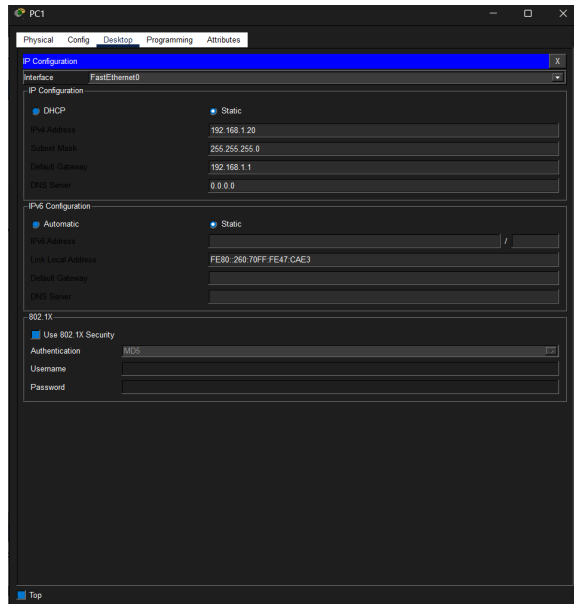
Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.20	255.255.255.0	192.168.1.1
Router	92.168.1.1	255.255.255.0	—

PC1 Configuration



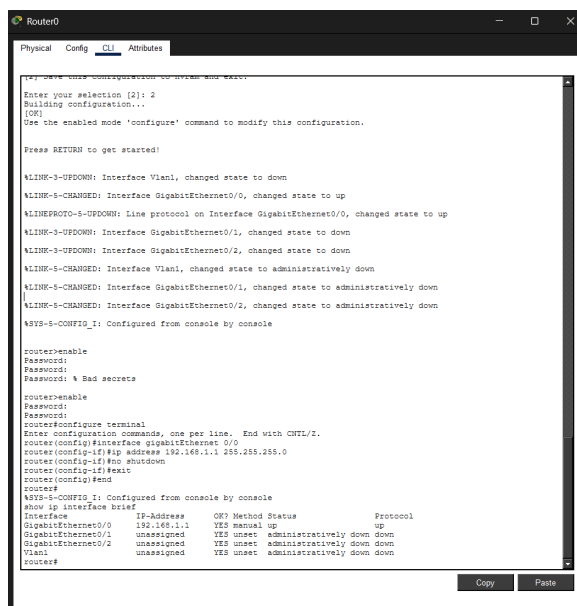
Configured the victim PC with IP address **192.168.1.10**, subnet mask **255.255.255.0**, and default gateway **192.168.1.1**.

PC2 Configuration



Configured the conceptual attacker PC with IP address **192.168.1.20**, subnet mask **255.255.255.0**, and default gateway **192.168.1.1**.

Router Configuration

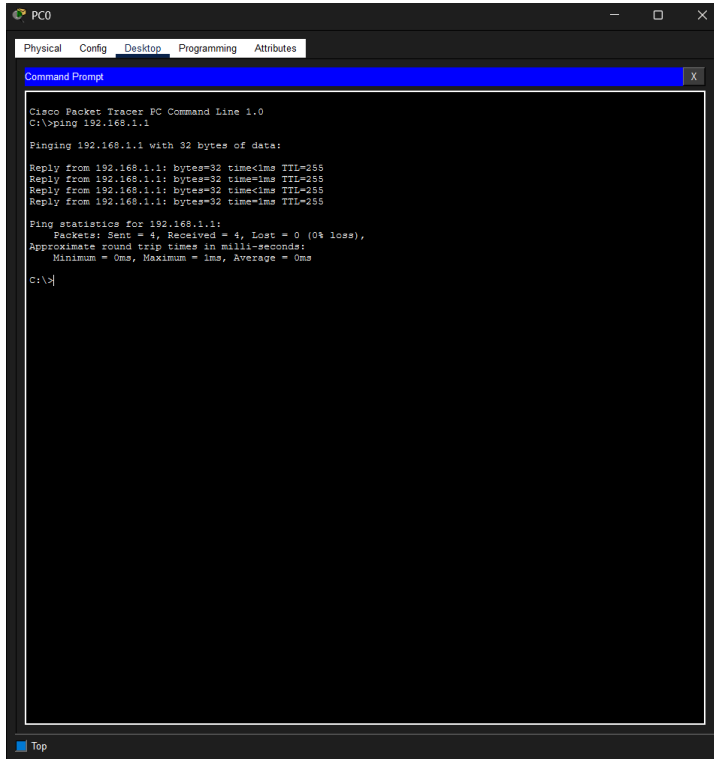


Configured the router interface with IP address **192.168.1.1** and verified that the interface was up and operational.

Step 3 - Test Normal Connectivity

From PC1, run:

ping 192.168.1.1



```
PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

Practical Observation:

PC1 successfully pinged the router 192.168.1.1, receiving 4 out of 4 replies with 0% packet loss. This confirmed normal connectivity between the victim PC, switch, and router before discussing the MITM concept.

Step 4 - Safe MITM Concept

The simulation demonstrates the conceptual position of an attacker between a victim and the intended destination. No real interception, ARP poisoning, credential capture, or traffic manipulation was performed.